

AE GATE Mentor - Problem Statement

The Challenge

GATE Aerospace Engineering aspirants often struggle with a vast syllabus, scattered learning resources, limited personalized guidance, and lack of instant doubt resolution. Students need an intelligent study assistant that provides concept explanations, numerical problem solving, quizzes, revision support, and progress tracking in one platform.

The Objective

Develop an AI-powered AE GATE Mentor using IBM watsonx.ai and Granite Models that delivers personalized study plans, GATE-level explanations, MCQs, formula sheets, analytics, and AI-powered mentoring through an interactive web application.

Core Features

- Personalized Study Planner
- AI Chat Mentor
- GATE-style Quiz Generator
- Formula & Revision Sheets
- Step-by-step Numerical Solutions
- Progress Analytics Dashboard
- IBM Granite-powered Responses